

Likelihood.

$$L(\lambda) = \prod_{i=1}^N \frac{e^{-\lambda} \lambda^{y_i}}{y_i!} = e^{-N\lambda} \lambda^{\sum_{i=1}^N y_i} \prod_{i=1}^N \frac{1}{y_i!}.$$

1. **Take logs.** (Use $\log(\prod a_i) = \sum \log a_i$, $\log(ab) = \log a + \log b$, and $\log(x^c) = c \log x$.)

$$\begin{aligned} \log L(\lambda) &= \sum_{i=1}^N [-\lambda + y_i \log \lambda - \log(y_i!)] \\ &= -N\lambda + \left(\sum_{i=1}^N y_i \right) \log \lambda - \sum_{i=1}^N \log(y_i!). \end{aligned}$$

2. **Differentiate $\log L(\lambda)$ with respect to λ .**

$$\begin{aligned} \frac{d \log L(\lambda)}{d\lambda} &= \frac{d}{d\lambda} [-N\lambda] + \frac{d}{d\lambda} \left[\left(\sum_{i=1}^N y_i \right) \log \lambda \right] - \frac{d}{d\lambda} \left[\sum_{i=1}^N \log(y_i!) \right] \\ &= -N + \left(\sum_{i=1}^N y_i \right) \frac{1}{\lambda} + 0 \\ &= -N + \frac{\sum_{i=1}^N y_i}{\lambda}. \end{aligned}$$

3. **Set the derivative to zero at $\lambda = \hat{\lambda}$ and solve.**

$$\left. \frac{d \log L(\lambda)}{d\lambda} \right|_{\lambda=\hat{\lambda}} = -N + \frac{\sum_{i=1}^N y_i}{\hat{\lambda}} = 0.$$

$$\frac{\sum_{i=1}^N y_i}{\hat{\lambda}} = N \iff \sum_{i=1}^N y_i = N\hat{\lambda} \implies \hat{\lambda} = \frac{1}{N} \sum_{i=1}^N y_i = \bar{y}.$$

4. **Second-order condition (concavity).** Need $\frac{d^2 \log L(\lambda)}{d\lambda^2} < 0$.

$$\frac{d^2 \log L(\lambda)}{d\lambda^2} = -\frac{\sum_{i=1}^N y_i}{\lambda^2} < 0 \quad (\lambda > 0 \text{ and } \sum_{i=1}^N y_i > 0).$$

Thus $\log L(\lambda)$ is strictly concave and the critical point $\hat{\lambda} = \bar{y}$ is the unique global maximizer.